

SATVIK PRASAD

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EDUCATION

Georgia Institute of Technology

B.S. in CmpE + Mathematics, 4.0 GPA

Atlanta, GA

Aug. 2025 – May 2029

- General projects team lead for 'TerraTrends' at Big Data Big Impacts @ GT
- FinancialAI Project Team at AI@GT, sponsored by Jane Street. Researching RL approaches to slippage optimization problems.
- Researching toroidal manifolds (torformer) for expressive, hierarchical self-attention in language modeling tasks under the mentorship of Emile Anand, affiliated with the ACO lab. Using nanochat as a harness, targeting ICLR 2027.
- John Martinson Honors Program — GT Venture Capital Club — Trading@GT.

Normanhurst Boys High School

Valedictorian, 4.0 GPA (Converted from ATAR)

Sydney, NSW, Australia

Feb. 2019 – Oct. 2024

EXPERIENCE

Cerebras Systems

Codesign / Nextgen Performance Intern

May 2026 - Aug 2026

Sunnyvale, CA

- Working on perf correlations and modeling for the next generation wafer-scale architecture.
- Modeling a FlashAttention-inspired softmax variant from scratch, achieved a ~ 1.01 correlation ratio w.r.t simulated kernels.
- Improved correlations between cycle-accurate sims, analytical perfmodels, and RTL simulators for critical kernels by an average of 189% (average ratio of Δ -to-RTL cycles before/after, activation-activation MatMuls, Softmax, Topk, FastPow2Exp, etc.)
- Modeling the mapping of a new compute-bound binary tree AllReduce variant used in TopK and FlashAttention kernels for our next-gen microarchitecture.

Cognito Tuition

Maths Ext. 1 & 2 Tutor and Educational Operations

Feb 2025 - Aug 2025

Sydney, Aus

- Taught 4-6 person classes, specialised in divide-and-conquer mathematical problem solving, designed cutting-edge homework and exam resources in LaTeX.

PROJECTS

tiny | Tiny ONNX inference engine in Rust. Optimised for consumer devices (0 runtime dependencies).

Dec 2025 - Current

- Implemented an ONNX parser using protobuf (prost.rs), using buffer liveness analysis and graph coloring to reduce memory footprint for lightweight models by $\sim 1.5x$.
- Achieved end-to-end MNIST inference with a $\sim 98KB$ total memory footprint (0.46 ms / inference), and a 482KB binary size.
- Implemented Conv2D, MaxPool, MatMul, and broadcast Add operations with appropriate kernel fusion, tiling, and strided access strategies for cache-friendly access.
- Designed zero-copy tensor views backed by a single arena allocation, avoiding per-op heap allocations entirely.

Promptly | Prompt-Based Agent with System-Wide Control — Ergo Challenge @ HackGT25

Sep 2025

- Used Omni-Parser pre-trained weights and RICO dataset to train a lightweight, local YOLO11n object detection model with 110x faster inference times.
- Utilised OpenCV for 50ms zero-shot text detection + pyautogui for system-wide control, designed a feedback-driven LLM event loop for automatic task correction.
- Outperforms Gemini 2.5 Computer Use Model inference latency by 10.5x.

Peggiator | Web-based, High Performance & Extensible Music Visualiser

Jan 2025 – May 2025

- 4x speedup compared with Chrome V8, used a self-derived WASM implementation of the Fast-Fourier Transform (radix-4), SIMD & Web Workers for optimal cache access.
- Captured system audio using a helper-binary written in Swift and ScreenCaptureKit, injected audio data into a Vite + React sidebar component for the DOM.
- Used Zig and WebGL for 3D rendering of manipulated audio waveforms, wrote custom affine transformations and GLSL shaders to minimise memory bottlenecks.

AWARDS

- Winner, CalHacks 12.0 @ UC Berkeley (2000+ competitors), ATAR 99.95 (Top 50 in Australia), Honorable Mention, Intercollegiate Math Tournament @ Columbia
- State Rank 10th in NSW for Mathematics Ext. 2, Silver Medal AIO, UNSW Scientia Scholar + Best Maths Student in NSW Award

TECHNICAL SKILLS

Languages: Rust, C/C++, Python, Typescript, Zig, OCaml, SQL, WebAssembly

Frameworks: React, Node.js, Express.js, FastMCP

Developer Tools: Git, Docker, AWS, NeoVim, Valgrind

Libraries: PyTorch, TensorFlow, OpenGL/WebGL, NumPy